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CHE 321; Spring - 2020; Quiz - 4

100 Points

A reversible reaction, $A + B \leftrightarrow C + D$ ($\Delta H_{rxn} = -100$ KJ/mol), is taking place in a CSTR at adiabatic conditions. The rate law for the reaction is:

$$-r_A = k \left(C_A C_B - \frac{1}{K} C_C C_D \right)$$

$$K_0 = 3.5673; H_1/R = 600.32 \text{ K}$$

$$K = K_0 \exp\left(\frac{H_1}{RT}\right) \rightarrow K = 3.5673 \exp\left(\frac{600.32}{T}\right)$$

1. If for a particular run in the CSTR, the conversion is found to be 0.8, what can you say about the maximum exit temperature possible for the CSTR? [50]

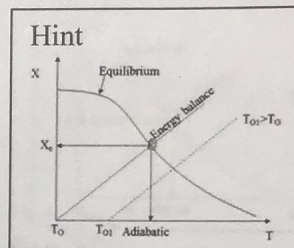
@ equilibrium

$$K = \frac{C_C C_D}{C_A C_B} = \frac{(0.8)(0.8)}{(0.2)(0.2)} = 16$$

$$16 = 3.5673 \exp\left(\frac{600.32}{T_{max}}\right)$$

$$T_{max} = 400.21 \text{ K}$$

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2. For a CSTR with heat exchanger (typical for the reaction above) in which the above reaction is being carried out, the following equation represents the energy balance. [25]

$$\underbrace{\dot{Q}}_1 + \underbrace{[-\Delta H_{Rx}^0(T_R) - \Delta \hat{C}_p(T - T_R)]F_{A0}X}_{2} + \underbrace{(-F_{A0}) \sum_{i=1}^n \Theta_i \tilde{C}_{pi}(T - T_{i0})}_{3} = 0$$

This equation has been divided into three parts. Which of these parts represent heat added to the system and which of these represent heat removed from the system? [25]

Heat Added (positive quantities) = 1, 2

Heat Removed (negative quantities) = 3

3. For the reaction in question #1, the reactants enter the CSTR at 275 K and the CSTR is run adiabatically. If $F_{A0} = F_{B0} = 8$ mol/s, $C_{PA} = C_{PB} = C_{PC} = C_{PD} = 400$ J/(mol $\cdot$ °C), and final temperature is 375 K, find the value of the heat change due to convection/flow. [25]

$$\begin{aligned} (100 \text{ kJ/mol}) \dot{Q}_{conv} &= - (F_{A0}) \sum_{i=1}^n \Theta_i \tilde{C}_{pi} (T - T_{i0}) = 0 \\ &= (8 \text{ mol/s}) (400) (375 - 275) \\ &= 320 \text{ kW} \end{aligned}$$

4. What is the conversion for problem #3?

$$(100 \text{ kJ/mol}) (8 \text{ mol/s}) (x) - 320 \text{ kW} = 0$$

$$x = 0.4$$